HARSH SINGHAL

AI/ML Engineer • Generative AI & Agentic Systems Architect

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PROFESSIONAL SUMMARY

AI/ML Engineer and CSE undergraduate specialized in Multi-Agent Systems, Agentic RAG, and Large Language Model (LLM) Orchestration. Experienced in designing decoupled, asynchronous full-stack architectures (FastAPI, React 18, LangGraph) integrated with probabilistic cognitive modeling (Continuous Bayesian Knowledge Tracing). Proficient in vector embeddings, dense semantic retrieval (ChromaDB), and PyTorch Transformer models. Strong algorithmic foundation with 320+ LeetCode problems solved.

EDUCATION

Ajay Kumar Garg Engineering College — *Bachelor of Technology in Computer Science & Engineering* **Oct. 2023 – May 2027**

Ghaziabad, UP, India

KEY ENGINEERING PROJECTS

SAGE: Autonomous Multi-Agent Socratic AI Tutor & Cognitive Engine | *LangGraph, FastAPI, React 18, ChromaDB, OpenAI, Vite, BKT* **Code • Live Demo**

- **Multi-Agent Orchestration:** Architected a stateful cyclic multi-agent graph with 7 specialized LangGraph agents (Curriculum Architect, Diagnostic Agent, Pedagogical Planner, SAGE Tutor, Assessor, Critic Quality Gate) enforcing strict Socratic pedagogy with zero direct-answer leakage.
- **Continuous Bayesian Knowledge Tracing:** Formulated real-time probabilistic cognitive modeling with slip (10%) and guess (20%) tolerance, dynamically updating latent student mastery $P(L_t)$ across prerequisite concept dependency DAGs.
- **Agentic Pedagogical RAG:** Engineered dense vector retrieval using persistent ChromaDB and `text-embedding-3-small` to dynamically inject verified academic textbook citations with normalized similarity scores, completely grounding inquiries.
- **Production Full-Stack Architecture:** Built an asynchronous FastAPI REST API with CORS middleware paired with a responsive React 18 SPA featuring zero-token topic initialization, real-time BKT meters, and live SVG Mermaid.js DAG rendering.
- **Quantitative Learning Analytics:** Evaluated student outcomes via Hake's Normalized Learning Gain ($g = (Post - Pre) / (1 - Pre)$), scientifically verifying pedagogy transition from baseline diagnostic to mastery ($g \geq 0.70$).

Agentic Document Intelligence & Multi-Tool Assistant | *LangGraph, ChromaDB, SQLite, Streamlit, Python, DuckDuckGo Search* **Code • Live Demo**

- **Stateful ReAct & Corrective RAG (CRAG):** Architected a stateful cyclic LangGraph agent with dynamic tool orchestration (Thread-Scoped RAG, DuckDuckGo search, Python arithmetic); engineered an autonomous CRAG fallback to web search when document context is insufficient.
- **Thread-Isolated RAG Engine:** Architected high-speed digital document ingestion (`PyPDFLoader`) with semantic chunking and persistent ChromaDB vector search; applied metadata filtering to guarantee 100% thread isolation and prevent cross-session document leakage.
- **RAG Triad Pipeline Evaluation:** Built an automated LLM-as-a-judge evaluation benchmark (`evaluator.py`) measuring Faithfulness, Answer Relevance, and Context Precision to systematically quantify grounding and eliminate hallucinations.
- **Interactive Streaming UI & Checkpointing:** Integrated LangGraph's `SqliteSaver` checkpointer for persistent multi-turn session memory; built a Streamlit SPA featuring token streaming, dynamic tool badges, and traceable page-level source citations.

Autoregressive Language Model from Scratch | *Python, PyTorch, Transformers, Deep Learning, NLP* **Source Code**

- Constructed and trained a decoder-only Transformer language model from scratch in PyTorch for autoregressive next-token prediction.
- Implemented multi-head self-attention mechanisms, causal masking to prevent future-token visibility, and learned positional encodings.
- Optimized training with Adam optimizer and Cross-Entropy loss; implemented temperature scaling and top-k sampling for diverse text generation.

TECHNICAL SKILLS

Languages: Python (Advanced), C/C++, JavaScript (ES6+), SQL, HTML5/CSS3

AI / Agentic & LLMs: LangGraph (Stateful Cyclic Graphs, Routing, Checkpointing), LangChain, Multi-Agent Coordination, RAG, Prompt Engineering, Model Context Protocol (MCP)

Machine Learning & Deep Learning: PyTorch, Transformers, Self-Attention, Neural Networks, NLP, Scikit-learn, OpenCV

Backend & Full-Stack: FastAPI, Uvicorn, Asynchronous REST APIs, React 18, Vite, Tailwind CSS, Streamlit, Pydantic v2

Databases & Vector Stores: ChromaDB, Persistent Vector Embeddings, Cosine Similarity Scoring, SQLite, PostgreSQL

CS Fundamentals & Tools: Data Structures & Algorithms, OOP, DBMS, OS, Computer Networks, Git, GitHub, Docker, Postman

ACHIEVEMENTS & COMPETITIVE MILESTONES

- **LeetCode (320+ Problems Solved):** Solved 320+ algorithmic challenges including 150+ Medium and 30+ Hard problems; demonstrated deep proficiency in Dynamic Programming, Graph Theory, and Tree Traversal algorithms.
- **Open-Source Architecture (SAGE):** Architected and developed a state-of-the-art educational AI platform combining cyclic LangGraph agents with Continuous Bayesian Knowledge Tracing and real-time learning analytics.
- **Academic Excellence:** Maintained strong academic standing in B.Tech Computer Science & Engineering while developing enterprise-grade AI applications and contributing to the AI developer ecosystem.